

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	28 September 2026
Team ID	6ab2210011b716a3e52c62c6
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The technical architecture of the project is designed using the ServiceNow Cloud Platform to automate the standard laptop procurement process. The user submits a laptop request through the ServiceNow Service Catalog. A Flow Designer workflow is triggered when the request is submitted.

The Flow Designer processes the request and uses the Create Catalog Task action to automatically create a procurement task. The Requested Item Record is used to pass the request information to the task. The request and task information are stored within the ServiceNow platform.

The complete application is hosted on the ServiceNow Cloud Platform, which provides the application environment and data storage. The workflow reduces manual processing and provides a standardized way to handle laptop orders.

Architecture Flow:

IT User → ServiceNow Service Catalog → Flow Designer → Create Catalog Task → Requested Item Record → ServiceNow Cloud Platform → IT Procurement Team

Streamlining IT Procurement

Automating Standard Laptop Orders with Flow Designer

IT User / Employee
Submits Standard Laptop Request



ServiceNow Service Catalog
Laptop Request Form



Flow Designer
Automation & Application Logic
Service Catalog Trigger



Create Catalog Task
Creates Procurement Task



Requested Item Record
Request Data / Task Data



ServiceNow Cloud Platform
Application Hosting + Data Storage



IT Procurement Team
Processes Standard Laptop Order

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Users submit standard laptop requests through the Service Catalog	ServiceNow Service Catalog
2	Application Logic-1	Captures and processes the laptop request	ServiceNow Flow Designer
3	Application Logic-2	Creates a catalog task for IT procurement	Flow Designer – Create Catalog Task
4	Application Logic-3	Uses requested item information to process the task	ServiceNow Request Item Record
5	Database	Stores requests, users, catalog items and task records	ServiceNow Database
6	Cloud Database	Cloud-based data storage used by ServiceNow	ServiceNow Cloud Platform
7	File Storage	Stores related documents or attachments if required	ServiceNow Attachments / File Storage
8	External API-1	No external API is required for the standard laptop ordering workflow	Not Applicable
9	External API-2	No second external API is required	Not Applicable
10	Machine Learning Model	Machine learning is not required for standard laptop order automation	Not Applicable

11	Infrastructure (Server / Cloud)	Hosts the application and automation workflow	ServiceNow Cloud Platform
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Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source framework is required for this ServiceNow-based project	Not Applicable
2	Security	User authentication, role-based access and ServiceNow platform security are used	ServiceNow Security / Access Controls
3	Scalable Architecture	The workflow can handle multiple standard laptop requests through automated Flow Designer processes	ServiceNow Flow Designer
4	Availability	The application is hosted on the ServiceNow cloud platform and can be accessed through the web	ServiceNow Cloud Platform
5	Performance	Automation reduces manual processing and creates procurement tasks automatically	Flow Designer